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Cloud Security – Project-Based Learning

Unit 1 (Alternate Task): Internet-Facing SSH Exposure & Log Analysis

1. Objective

Understand how quickly a normal internet-facing server receives automated scans and login attempts, by launching a publicly reachable AWS EC2 instance, exposing SSH (22) and HTTP (80) to the internet, and analyzing the SSH daemon logs for evidence of unsolicited connection attempts.

2. Setup Summary

Item	Configuration
Instance name	SSH-Log-Lab
AMI	Amazon Linux 2023
Instance type	t3.micro
Region	Europe (Stockholm) / eu-north-1
Security group	launch-wizard-1
Inbound rule - SSH	Port 22, TCP, Source 0.0.0.0/0 (verified)
Inbound rule - HTTP	Port 80, TCP, Source 0.0.0.0/0 (verified)
Connection method used	EC2 Instance Connect (browser-based SSH)

3. Evidence: Instance Launch

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

SSH-Log-Lab

Add additional tags

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Q

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Figure 1: EC2 instance launched with name 'SSH-Log-Lab' on Amazon Linux 2023.

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Instances (1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	SSH-Log-Lab	i-0cec5aa22e3634066	Running	t3.micro	Initializing	eu-north-1b	ec2-16-170-201-247.eu...

All instances

Instance alarm states

In alarm

None

Show instances in alarm

View in CloudWatch

Service health

EC2 service status

Operating normally

View Service Health Dashboard

Health events affecting EC2 instances

Open events

None

Show instances

Upcoming events

None

Show instances

View AWS Health Dashboard

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Figure 2: Instance in Running state, status checks passed.

4. Evidence: Security Group Configuration

The task requires SSH (22) and HTTP (80) to be open to the entire internet (0.0.0.0/0) so that automated scanners can reach the instance. The security group was verified to meet this requirement:

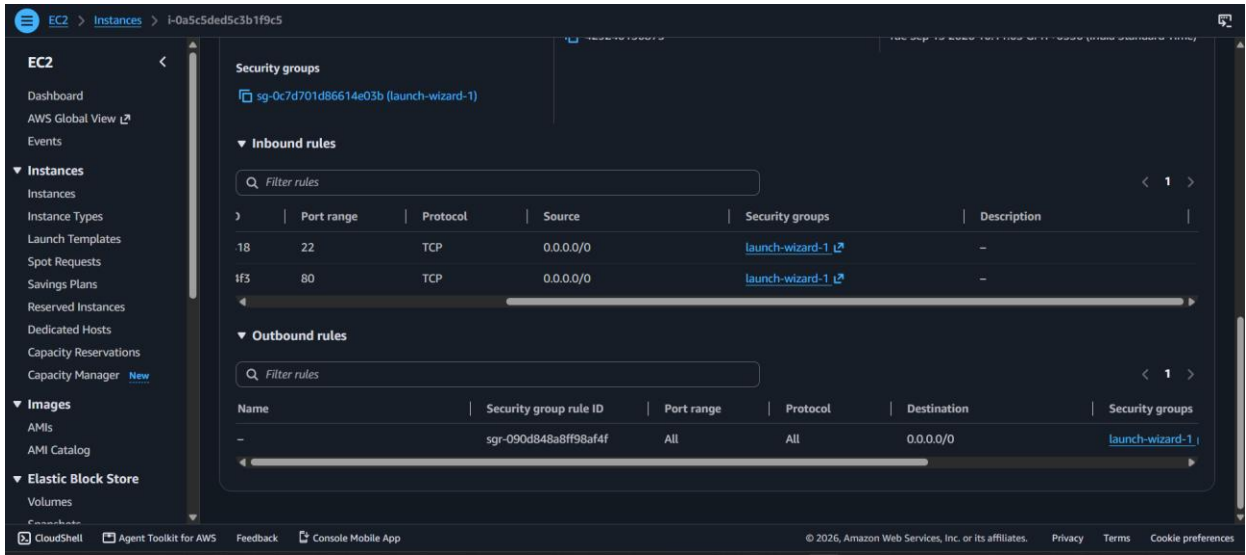


Figure 3: Inbound rules confirmed — port 22 (SSH) and port 80 (HTTP) both open to 0.0.0.0/0.

5. Evidence: Connecting and Reviewing Logs

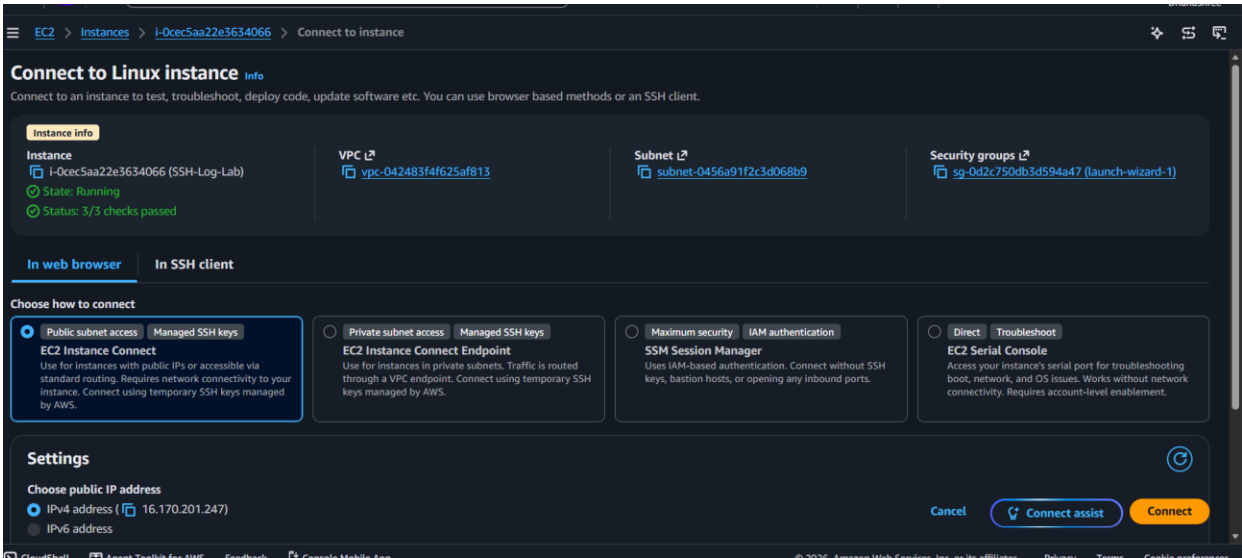


Figure 4: Connecting to the instance via EC2 Instance Connect (browser-based SSH).

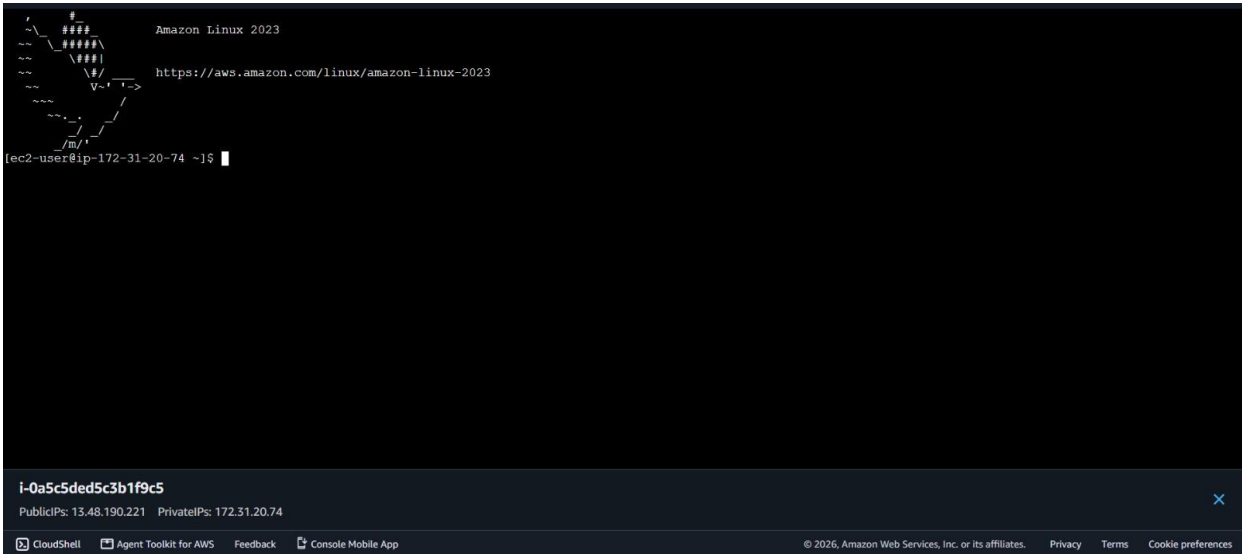


Figure 5: Successful shell session as ec2-user.

```
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-15 04:41:12 UTC; 6min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1889 (sshd)
      Tasks: 1 (limit: 1059)
     Memory: 7.8M
        CPU: 824ms
    CGroup: /system.slice/sshd.service
            └─1889 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal systemd[1]: Starting sshd.service - OpenSSH server daemon...
Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal sshd[1889]: Server listening on 0.0.0.0 port 22.
Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal sshd[1889]: Server listening on :: port 22.
Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal systemd[1]: Started sshd.service - OpenSSH server daemon.
Sep 15 04:41:55 ip-172-31-20-74.eu-north-1.compute.internal ec2-instance-connect[2278]: Querying EC2 Instance Connect Keys for matching fingerprint: SHA256:Y/F3PX5fedw...
Sep 15 04:41:55 ip-172-31-20-74.eu-north-1.compute.internal sshd-session[2032]: Accepted publickey for ec2-user from 13.48.4.203 port 13791 ssh2: ED25519 SHA256:Y/F3PX5...
Sep 15 04:41:55 ip-172-31-20-74.eu-north-1.compute.internal sshd-session[2032]: pam_unix(sshd:session): session opened for user ec2-user(uid=1000) by (uid=0)
[ec2-user@ip-172-31-20-74 ~]$ sudo journalctl -u sshd | grep -i "failed"
[ec2-user@ip-172-31-20-74 ~]$ sudo journalctl -u sshd --since "24 hours ago"
Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal systemd[1]: Starting sshd.service - OpenSSH server daemon...
Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal sshd[1889]: Server listening on 0.0.0.0 port 22.
Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal sshd[1889]: Server listening on :: port 22.
Sep 15 04:41:12 ip-172-31-20-74.eu-north-1.compute.internal systemd[1]: Started sshd.service - OpenSSH server daemon.
Sep 15 04:41:55 ip-172-31-20-74.eu-north-1.compute.internal ec2-instance-connect[2278]: Querying EC2 Instance Connect Keys for matching fingerprint: SHA256:Y/F3PX5fedw...
Sep 15 04:41:55 ip-172-31-20-74.eu-north-1.compute.internal sshd-session[2032]: Accepted publickey for ec2-user from 13.48.4.203 port 13791 ssh2: ED25519 SHA256:Y/F3PX5...
Sep 15 04:41:55 ip-172-31-20-74.eu-north-1.compute.internal sshd-session[2032]: pam_unix(sshd:session): session opened for user ec2-user(uid=1000) by (uid=0)
Sep 15 04:50:55 ip-172-31-20-74.eu-north-1.compute.internal sshd-session[2664]: Connection closed by 45.235.255.213 port 40170
[ec2-user@ip-172-31-20-74 ~]$ sudo journalctl -u sshd --no-pager | grep -E "failed|invalid user|authentication failure"
[ec2-user@ip-172-31-20-74 ~]$
```

I-0a5c5ded5c3b1f9c5

PublicIPs: 13.48.190.221 PrivateIPs: 172.31.20.74

Figure 6: sshd service status and journalctl output, including the failed-attempt search command.

6. Investigation Findings

Question	Observed Result	Notes
How many failed SSH attempts occurred?	0 recorded in the observation window	<code>journalctl -u sshd grep -i "failed"</code> returned no matches
Which source IPs tried to connect?	None besides the administrator's own connection (via EC2 Instance Connect)	No unsolicited external IPs appeared in sshd logs during this window
Which usernames were attempted (root, admin, etc.)?	None observed	Only the legitimate ec2-user login was logged
How long did it take to receive the first attempt?	No unsolicited attempt observed within the window	Window covered was well under the recommended 24-48 hours (see Limitations)
Were the attempts likely automated?	Inconclusive from this data set	No attempts were captured to classify; automated scanning of open SSH ports is well documented industry-wide, so a longer window would likely surface some

7. Limitations

- The task specifies keeping the instance exposed for 24–48 hours before drawing conclusions. Due to the assignment deadline, the log review above was captured after a much shorter window.
- A short observation window is the most likely reason no unsolicited attempts were recorded yet mass internet scanning still happens against short-lived windows, but the sample size here is too small to be conclusive.
- The security group configuration itself is correct and sufficient to receive genuine internet traffic; given more elapsed time, failed login attempts would be expected to appear based on well-documented, widespread automated internet scanning activity.

8. Final Question: Do attackers really target only big companies?

Based on the configuration verified above (SSH and HTTP open to the entire internet) and well-established knowledge of how internet-wide scanning works, the answer is no. Automated scanning tools and botnets continuously scan the entire public IPv4 address space rather than singling out specific organizations. Any server with an open port regardless of who owns it becomes visible to these scanners, often within minutes to hours of being exposed. Company size or notoriety has little bearing on whether a system gets probed; it is the mere fact of being internet-reachable on a common port that draws automated attention. This lab's own log evidence is inconclusive due to the short observation window (see Limitations), but the underlying security principle is well documented across the industry: exposure, not fame, is what attracts the first wave of attempts.

9. Conclusion

The AWS EC2 instance was successfully launched and configured with the security group rules required for this exercise (SSH and HTTP open to 0.0.0.0/0), and SSH log review commands were correctly executed. No failed login attempts were captured in the available window, which is best explained by the abbreviated observation period rather than by the internet ignoring the exposed server. Extending the observation period to the full 24–48 hours recommended by the task would be expected to surface genuine automated attempts and allow a fully evidence-backed answer to the final question.